# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Information Technology

## CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

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|---|---|---|---|
| **Course Code:** | CSA10 / CSA1063 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop** |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO1 Statement:** | *Apply advanced methodologies and agile project management techniques to manage software projects effectively* | | |
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| **Date:** | 27-07-2026 | **Duration:** | 60 minutes |

**Code of Conduct**

*I Jasim Asmaan A / 192421060 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: A.Jasim Asmaan

## Annexure A – Agile Sprint Planning & User Story Workshop

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.

### Project Vision

The **AI-Based Smart Disaster Alert and Emergency Response Coordination System** is designed to provide early warnings for natural disasters using artificial intelligence and real-time monitoring. The system analyzes weather conditions, sensor data, and location information to predict disaster severity and send timely alerts to people in affected areas. It also helps government authorities and rescue teams coordinate emergency operations by providing nearby shelter details, food and first-aid information, emergency contact numbers, and live disaster updates.

The project aims to build a secure, reliable, scalable, and user-friendly platform that improves disaster preparedness, reduces the loss of lives and property, and enables faster emergency response during critical situations.

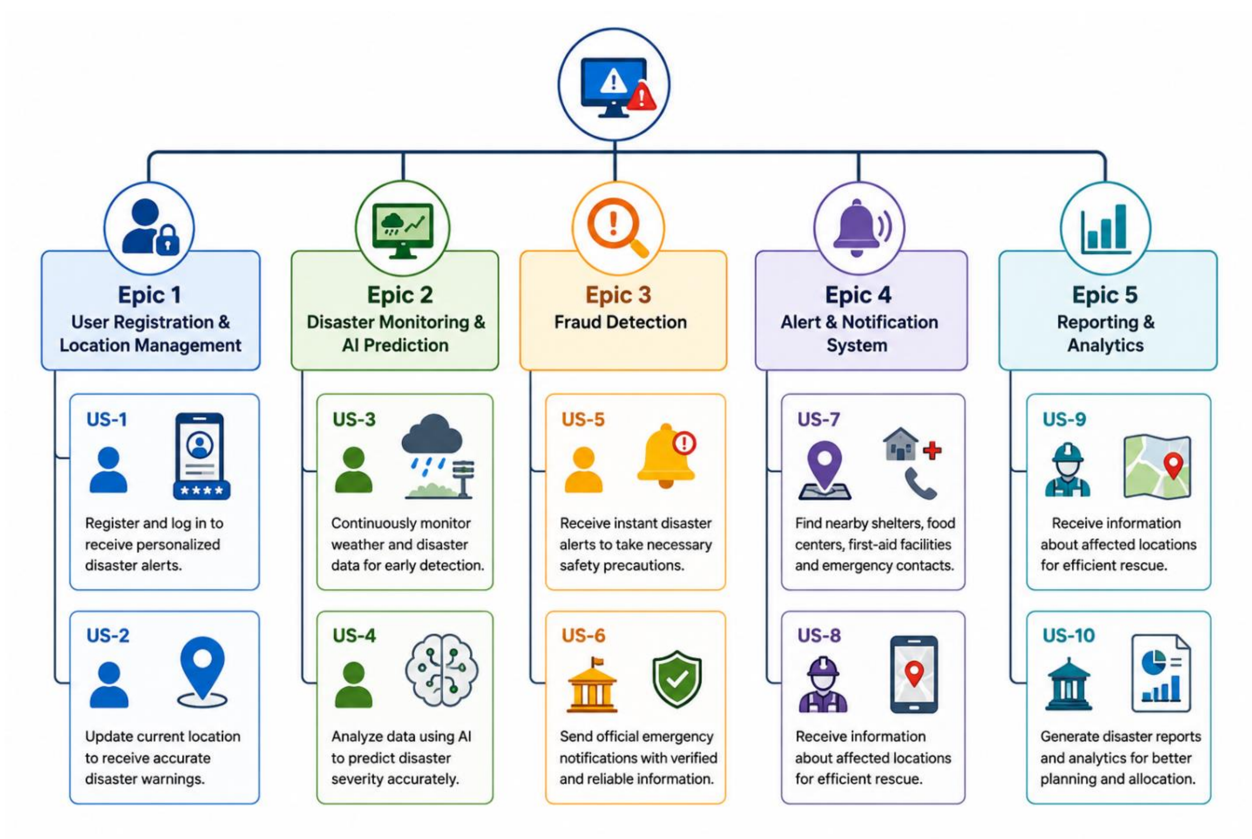
## Project Objectives

- Predict natural disasters using AI and real-time monitoring.
- Send instant location-based disaster alerts to citizens.
- Improve coordination between rescue teams and government authorities.
- Provide nearby shelter, food, first-aid, and emergency contact information.

| Stakeholder | Roles and Responsibilities |
|---|---|
| **Citizens** | Receive alerts and safety instructions. |
| **Disaster Management Authority** | Monitors disasters and coordinates rescue operations. |
| **Rescue Teams** | Rescue affected people and provide emergency assistance. |
| **Hospitals** | Provide medical treatment during emergencies. |
| **Police & Fire Department** | Maintain public safety and assist rescue operations. |
| **Weather Department** | Provides weather and disaster data. |
| **Government Authorities** | Manage disaster response and public safety. |
| **System Administrator** | Maintains the software and database. |

2. **Create Epics and User Stories**

- ❖ Break down the requirements into Epics and formulate well-structured User Stories using the format:
  *As a <user>, I want <feature>, so that <benefit>.*




## Epic 1 – User Registration & Location Management
### User Story 1

As a citizen, I want to register and log in to the application so that I can receive personalized disaster alerts based on my location.

### User Story 2

As a citizen, I want to update my current location so that I can receive accurate disaster warnings for my area.

## Epic 2 – Disaster Monitoring & AI Prediction

### User Story 3

As a disaster management officer, I want the system to continuously monitor weather and disaster data so that disasters can be detected at an early stage.

### ‘User Story 4

As an AI prediction system, I want to analyze weather patterns and sensor data so that I can predict the disaster severity level accurately.

## Epic 3 – Emergency Alert & Notification System

### User Story 5

As a citizen, I want to receive instant disaster alerts so that I can take necessary safety precautions immediately.

### User Story 6

As a government authority, I want to send official emergency notifications so that citizens receive only verified and reliable information.

## Epic 4 – Emergency Alert & Notification System

### User Story 7

As a citizen, I want to find nearby shelters, food centers, first-aid facilities, and emergency contact numbers so that I can get help quickly during emergencies.

### User Story 8

As a rescue team member, I want to receive information about affected locations so that rescue operations can be carried out efficiently.

## Epic 5 – Reporting & Administration

### User Story 9

As a rescue team member, I want to receive information about affected locations so that rescue operations can be carried out efficiently.

### User Story 10

As a government official, I want to generate disaster reports and analytics so that future disaster planning and resource allocation can be improved.

3. **Define Acceptance Criteria**

- ❖ Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

## User Story 1

- **Given** the citizen has successfully registered an account,
- **When** the citizen logs into the system
- **Then** the system should provide secure access and display the user dashboard.

## User Story 3

- **Given** real-time weather and sensor data are available
- **When** the system analyzes the collected data
- **Then** it should predict the disaster severity level accurately**.**

## User Story 5

- **Given** a disaster is detected in a particular area,
- **When** the disaster risk reaches the warning level,
- **Then** the system should send instant alerts to all affected users based on their location.

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## User Story 7

- **Given** a citizen requests emergency assistance,
- **When** the system identifies the user's current location,
- **Then** it should display nearby shelters, hospitals, first-aid centers, emergency contact numbers, and government support information.

4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

| Priority | Product Backlog Item | Business Value |
|---|---|---|
| High | User Registration & Authentication | Very High |
| High | Real-Time Disaster Monitoring | Very High |
| High | AI-Based Disaster Severity Prediction | Very High |
| High | Emergency Alert Notification | High |
| Medium | Nearby Shelter & First-Aid Finder | Medium |
| Medium | Emergency Contact & Government Information | Medium |
| Medium | Rescue Team Coordination | Medium |
| Low | Disaster Reports & Analytics | Low |
| Low | Admin Dashboard | Low |
| Low | System Settings | Low |

## Sprint Backlog (Sprint 1)

| User Story | Development Tasks | User Story |
|---|---|---|
| User Registration | Design registration page, user login, password encryption | User Registration |
| Disaster Monitoring | Collect weather and sensor data, monitor disaster conditions | Disaster Monitoring |
| AI Prediction | Implement AI model, predict disaster severity | AI Prediction |
| Alert System | Develop SMS, Email, and App notifications | Alert System |
| Shelter & Emergency Services | Integrate shelter database, emergency contacts, and first-aid information | Shelter & Emergency Services |

5. **Plan the Sprint Backlog**
    - Select high-priority user stories for the first sprint and divide them into actionable development tasks.

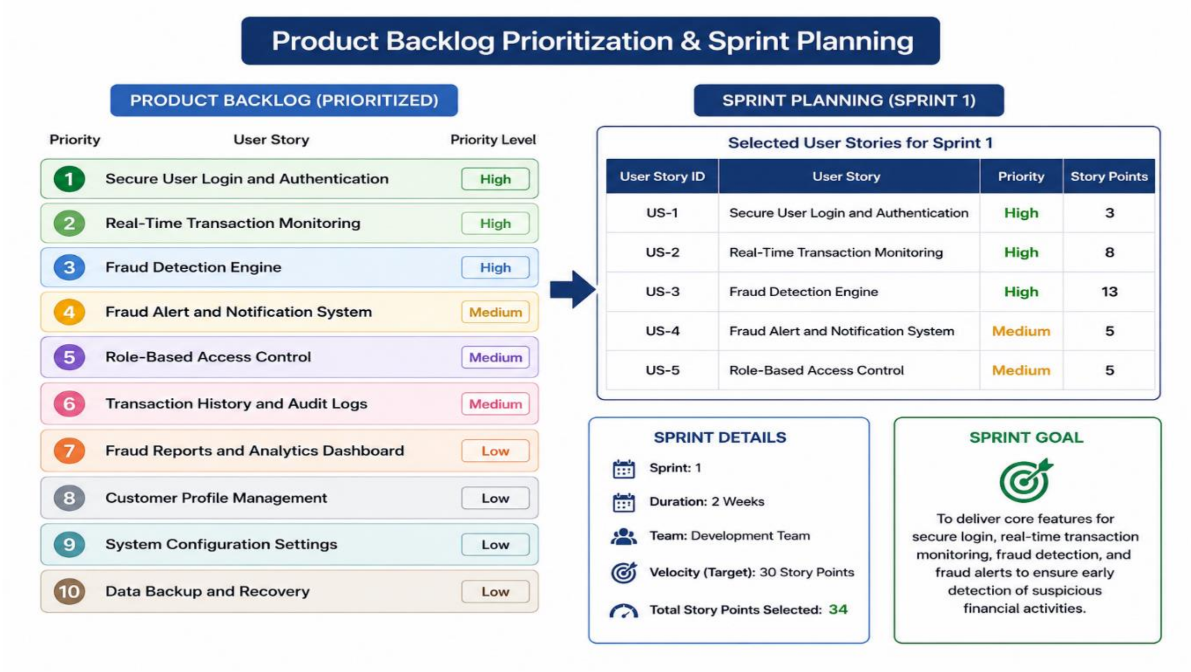
| User Story | Story Points | Justification |
|---|---|---|
| **User Registration & Login** | 3 | Simple user registration and secure login functionality. |
| **Real-Time Disaster Monitoring** | 8 | Requires continuous monitoring of weather and sensor data. |
| **AI-Based Disaster Severity Prediction** | 13 | Involves AI analysis, prediction models, and real-time decision making. |
| **Emergency Alert Notification** | 5 | Integrates SMS, app, and email notification services. |
| **Shelter & Emergency Services** | 5 | Displays nearby shelters, hospitals, emergency contacts, and first-aid information using location services. |

## Estimation Technique

### Fibonacci Sequence (1, 2, 3, 5, 8, 13)

Reason:

- Industry-standard Agile estimation technique.
- Helps estimate the complexity of each user story.
- Considers development effort and technical difficulty.
- Supports better sprint planning and task prioritization.
- Helps the development team distribute work efficiently.




6. **Estimate Effort Using Story Points**
    - ❖ Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

- **User Registration & Authentication – 3 Story Points**
    - Simple implementation of user registration and secure login.
    - Low complexity with basic authentication and validation.
- **Real-Time Disaster Monitoring – 8 Story Points**
    - Collects and monitors real-time weather and sensor data.
    - Requires continuous data processing and monitoring.
    - Higher complexity because of real-time information handling.
- **AI-Based Disaster Severity Prediction – 13 Story Points**
    - Uses AI to analyze disaster-related data.
    - Predicts disaster severity based on weather and location.
    - Most complex module due to AI processing and decision-making.
- **Emergency Alert Notification System – 5 Story Points**
    - Sends instant alerts through SMS, mobile app, and email..
    - Notifies users based on their current location
    - Provides disaster safety instructions and evacuation guidance..
- **Shelter & Emergency Services – 8 Story Points**
    - Displays nearby shelters, hospitals, food centers, and first-aid facilities..
    - Provides emergency contact numbers and government support information.
    - Requires GPS integration and location-based services..

## Justification for Story Point Estimation

- The Fibonacci sequence is widely used in Agile software development.
- It estimates the relative complexity of each user story rather than the actual development time.
- It helps prioritize high-value features during sprint planning.
- It improves workload distribution among team members.
- It considers implementation effort, technical complexity, uncertainty, and project risk.

7. **Present the Sprint Goal and Expected Deliverables**
    - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

## Sprint Goal

The Fibonacci sequence is a widely used estimation technique in Agile development.

- It measures the relative complexity of each user story instead of the exact development time.
- It helps the team prioritize important features during sprint planning.
- It improves workload distribution among team members.
- It considers implementation effort, technical complexity, uncertainty, and project risks.


## Sprint Objectives

- Sprint Objectives
- Develop a secure user registration and login system.
- Monitor disaster-related data in real time from reliable sources.
- Predict disaster severity using AI based on location and weather conditions.
- Send instant emergency alerts through SMS, mobile app, and email.
- Display nearby shelters, hospitals, emergency contacts, and first-aid information.
- Provide a simple dashboard for government authorities and administrators.
- Expected Sprint Deliverables

## Expected Sprint Deliverables

At the end of Sprint 1, the following features will be completed:

- Secure user registration and authentication.
- Real-time disaster monitoring and data collection.
- AI-based disaster severity prediction.
- Location-based emergency alert notification system.
- Display of nearby shelters, hospitals, food aid centers, and emergency contact details.
- Basic administrator dashboard to monitor disaster information and alerts.

- Secure storage of user and disaster-related data.
- A working prototype ready for testing.
- Basic administrator dashboard to monitor transactions.
- A working prototype ready for testing.